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Advanced statistical analysis of vortex-induced vibrations in suspension bridge hangers with and without Stockbridge dampers

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ABSTRACT

This paper presents a detailed statistical analysis of strong hanger vortex-induced vibrations (VIV) at the Hålogaland Bridge in Narvik, Norway. Severe VIV during construction led to the installation of Stockbridge dampers post-completion. Unfortunately, many dampers broke within a year, prompting a long-term measurement campaign. The measurements highlight the complexity of observed VIV, with non-stationary and multi-frequency vibrations during constant wind speeds. The paper assesses the effectiveness of various damper configurations on the hangers and finds that a single damper notably reduces vibration amplitudes, however, installing more dampers results in a lower observed difference. The research includes a detailed statistical analysis of wind data and cable responses, considering different observation intervals since the observed time of development of lock-in vibrations might impact the VIV statistical indicators. It is also shown that the duration over which wind can be considered stationary most often differs from the conventional 10-minute duration. Finally, using statistical hypothesis testing, it is demonstrated that VIV metrics are slightly influenced by the observation interval length, and it is confirmed that high turbulence intensity significantly limits the amplitude reached at synchronization. Overall, this research provides valuable insights into understanding and addressing challenges related to measuring and interpreting vortex-induced vibrations on hangers.

Introduction

The spans of long suspension bridges are progressively extending in length worldwide. To address the resulting structural integrity and dynamic behaviour concerns, innovative solutions are needed. As the span increases, the length of the hangers also increases proportionally. Over the years, there has been a significant evolution in hanger designs in suspension bridges. In older structures like the Golden Gate Bridge in the USA or the Forth Road Bridge in the UK, the hangers were typically made of stranded wire loops (Abdel-Ghaffar and Scalan, 1985). These loops were embedded into grooves on the cable bands and connected to the deck's top chord using threaded bolts. In contrast, more recent hanger designs utilize locked coil cables or parallel wire strands. These are often coated with a high-density polyethylene layer for protection (Jamal and Sundet, 2021; Larsen et al., 2022). The ends of these modern hangers are usually attached to the cable bands and deck anchor plates through fork socket/eye plate connections. These modern connections are characterized by significantly lower friction than their predecessors. Coupled with the characteristic lower damping ratios of super long hangers (Jafari et al., 2020), the hangers are more vulnerable to wind-induced vibrations. Several studies have investigated the interaction between flexible cables and fluid flow, examining the resulting dynamic behaviour. Fundamental examples can be found in the reviews on flow-induced vibrations of bluff bodies (Parkinson, 1989), which can be rigid or free to oscillate (Sarpkaya, 1979, 2004; Bearman, 1984). Whenever a bluff body, typically a circular cylinder is submerged in a fluid flow, it causes flow separations from its surface, and the separated flow creates an unsteady wake downstream of the object. The creation of these vortices arises from the roll-up of the unstable shear layers produced during flow separation. For sub-critical Reynolds numbers (from 300 to $2 \cdot 10^5$), this occurrence leads to an alternating pressure loading on the surface of the cylinder that creates a transverse force at the vortex-shedding frequency f_{vs}. For a rigid circular cylinder, this frequency is related to the well-known Strouhal number (Lienhard, 1966):

$$\mathrm{St} = \frac{f_{vs} D}{U} \tag{1}$$

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Here, U is the steady velocity of the fluid flow and D is the characteristic dimension of the body. The Strouhal number, for circular cylinders, is nearly constant and equal to 0.18 for Reynolds numbers in the subcritical range (Lienhard, 1966). Therefore, if left free to move, the body will vibrate in the cross-flow direction at a frequency equal to f_{vs}. This well-known phenomenon is called Vortex-Induced Vibration (VIV). Once the vortex-shedding frequency gets near to one of the natural frequencies of the body f_n, the phenomenon of *synchronization* or *lock-in* occurs wherein the structure undergoes near-resonance vibration, which can lead to large structural displacement. In lock-in conditions, the vortex shedding frequency deviates from the direct correlation with the Strouhal number and becomes predominantly influenced by the body's motion. This interaction between the structure and the airflow exhibits a highly nonlinear dynamic. Therefore, for small variations of the flow speed U, the shedding frequency remains "locked-in" to the structure vibration frequency f_n.

The phenomenon of lock-in is part of a nonlinear dynamic process characterized by a hysteresis loop. This aspect has been thoroughly documented in the scientific literature through forced and free vibration experiments (Feng, 1968; Bishop and Hassan, 1964). Sudden jumps in vibration amplitude characterize the observed hysteresis loop. The occurrence of the jumps is affected by whether the flow velocity is raised from lower speeds or reduced from higher speeds. The hysteresis loop could be attributed to a nonlinear stiffness or damping behaviour, thus modelling the phenomena as an elastic system (Staubli, 1981). However, based on the findings by Feng (1968), the work by Parkinson (Parkinson, 1989) proposes that hysteresis conditioning arises from the fluid mechanics, therefore from the lift forces, rather than in the elastic system. Several researchers have achieved considerable success in approximating experimental results by modelling the wake fluid system using a nonlinear wake oscillator in conjunction with the equation of motion for a mass–spring–damper system (Tamura, 2020; Facchinetti et al., 2004; Hartlen and Currie, 1970; Denoël, 2020). Beyond wake-oscillator models, which effectively approximate the entire lock-in range, the scientific literature also extensively features single-degree-of-freedom deterministic models. Well-known contributions in this area have been made by Fazl Ehsan and Scanlan (1990), and Vickery and Basu (1983b,a), who model the aerodynamic forces as a function of position and velocity. While single-degree-of-freedom models offer more precise approximations of the peak amplitude during lock-in, they require specific calibration for each distinct lock-in region, contrasting with the broader applicability of wake-oscillator models.

The amplitude of the cylinder response when vibrating during lock-in and the width of the lock-in region depends strongly on the ratio between the damping and excitation forces, which is usually expressed as a reduced damping parameter. One possible representation of this ratio used in literature (Gabbai and Benaroya, 2005) is the Scruton number:.

$$\mathrm{Sc} = \frac{4\pi\xi_s m}{\rho D^2} \tag{2}$$

where ξ_s is the structural damping ratio of the cylinder, m is the mass per unit length, ρ is the density of the fluid, and D is the diameter of the cylinder. As the Scruton number decreases, the lock-in is characterized by an increase in structural amplitude and occurs over a wider band of wind velocities (Gabbai and Benaroya, 2005; Bearman, 1984).

During lock-in, the vibration amplitude is usually in the order of fractions of D, and does not represent an immediate structural threat. However, this kind of periodic excitation often occurs at low wind speeds and frequent wind-induced excitation can be detrimental to the fatigue life of the structure. Therefore, it is often necessary to intervene with mitigation devices that change the system's properties, such as mass, damping, or stiffness, to prevent VIV from occurring.

In recent years, numerous field observations have been documented in the literature. Vortex-induced vibrations (VIV) have been observed in the hangers of the Xihoumen Bridge in China (Deng et al., 2021), the Akinada Bridge in Japan (Fujino et al., 2012), and the Gjemnessund Bridge in Norway (Hjorth-Hansen and Strømmen, 2003). Wake-induced vibration on hangers has also been reported by Hu et al. (2023), and on a guyed mast by Denoël and Andrianne (2017). Additionally, wake-induced vibration, along with rain–wind-induced vibration, has been observed in many cable-stayed bridges, such as the Fred Hartman Bridge in the USA (Zuo et al., 2004; Caetano, 2007), the Erasmus Bridge in the Netherlands (Caetano, 2007), and in cable-stayed structures like the pedestrian and bicycle roundabout in Eindhoven (Argentini et al., 2013). Matsumoto et al. (2003) conducted field experiments focused on rain–wind-induced vibration on cables, providing further insights into these phenomena.

Various techniques have been explored to mitigate VIV in these cases. The approaches generally fall into two categories: posterior damping strategies and preventive design guidelines. The former includes control mechanisms involving the installation of devices that add damping to the vibrating cable, thereby reducing the vibrations without altering or reducing the mechanism of wind loading on the cables. Classic tuned mass dampers (TMDs), consisting of an inertial mass linked to the cable by a linear spring and a damper, are a common solution (Lazar et al., 2016). More recent proposals include eddy-current TMDs (Niu et al., 2013) and TMDs coupled with elastic spacers between the hangers (Li et al., 2022). Another frequently used solution is a special kind of nonlinear hysteretic damper known as Stockbridge dampers, which are normally used in overhead power lines (Wagner et al., 1973; Barbieri and Barbieri, 2012; Markiewicz, 1995). Numerous studies in the literature focus on the experimental characterization and modelling of this device, using both analytical (Diana et al., 2003; Foti and Martinelli, 2018; Di et al., 2020) and numerical models (Luo et al., 2016).

The latter category encompasses methods to prevent VIV by modifying the structure's aerodynamic properties. These active or passive methods focus on controlling the flow detachment from the structure to reduce vortex formation in the cable wake. Common solutions include adding surface features like bumps or spiral wires to the cable (Gu and Du, 2004; Owen et al., 2001). Proposals for actively controlling wake flow have also been made, such as using suction flow control to suppress flow separation from the structure (Chen et al., 2013; Patil and Ng, 2012), or a combination of blowing and suction (Li et al., 2003; Fransson et al., 2004).

These vibration mitigation strategies have been extensively tested through experiments, numerical simulations, and field measurements. On cable-stayed bridges, Persoon and Noorlander (1999) tested the damping performance of polyethene ropes, while Kim et al. (2022) experimented with a Stockbridge damper configuration on stay cables during non-synoptic wind events. Gao et al. (2024) applied a similar technique on the hangers of a suspension bridge to monitor VIV and subsequently attempted to control the vibrations through the installation of special Stockbridge dampers. (Main and Jones, 2001) measured the performance of linear dampers on the Fred Hartman Bridge in the USA. In the context of suspension bridges, Deng et al. (2021) demonstrated that wake-induced vibration measured in the field could be replicated with wind tunnel experiments. Cantero et al. (2018) showed the feasibility of detecting VIV by monitoring vibrations in the bridge's deck.

However, providing a link between theoretical and experimental results to field data measurements is challenging, primarily due to the complexity of predicting turbulence's impact on shear layer instability (Sarpkaya, 2004; Fenerci et al., 2023). Some wake-oscillator models that include the effect of stochastic turbulence are present in the literature (Denoël, 2020; Mannini, 2020), but currently, such numerical models lack validation in real-life conditions. This complexity is further increased by uncertainties related to the limited number of sensors deployed in the field and to commonly adopted assumptions, such as wind field homogeneity, which may not hold for very long hangers. In addition, there is a lack of studies on the long-term performance of deployed Stockbridge dampers under a variety of operational conditions, extending beyond dynamic tests in controlled experimental settings.

Fig. 1. The Hålogaland bridge.

In addressing these challenges, this study focuses on field data collected from a long-term monitoring campaign on the Hålogaland Suspension Bridge. The first goal is to examine the performance of Stockbridge dampers, a commonly used option in mitigating VIV in cable bridges. The study compares different damper configurations installed on the three longest hangers of the bridge, providing practical insights into the effectiveness of mitigation strategies. By analysing data from both damped and undamped hangers, a deeper understanding of the complex dynamics involved in VIV is gained.

To enhance the observations, an advanced statistical analysis is employed on the response and the wind data. This includes testing the independence of the statistical indicators of the hangers' response from the observation window. Additionally, the influence of turbulence intensity on the maximum response induced by vortex shedding is explored by drawing insights directly from the measured data.

Essentially, this work provides an analysis of real-world field data, offering a practical exploration of VIV in the long cables of suspension bridges and its mitigation. By combining empirical evidence and statistical analysis, valuable information is presented for processing field data related to vortex-induced cable vibration.

1. The Hålogaland bridge

The Hålogaland bridge (Fig. 1) is a suspension bridge located in Narvik in Northern Norway. It was completed in 2018, becoming the longest suspension bridge above the Arctic Circle. As shown in Fig. 2, the direction of the bridge is almost aligned with the North–South direction, deviating from the magnetic North by only 1.1°. The surrounding topography is dominated by steep mountainsides and fjords. Immediately South of the bridge is Fagernesfjellet which is 1272 m high, shielding the bridge from southern winds. The main span of the bridge is 1145 m and the longest hangers are about 120 m. The deck is a closed steel box girder and it is 18.6 m wide, supporting two traffic lanes and a bicycle and pedestrian lane. The bridge has air-spun steel main cables and locked-coil steel hangers. Furthermore, the main cables lay in a slightly tilted plane because of the particular A-shaped towers. This causes the hangers to be slightly inclined in the vertical plane. The inclination goes from 2.8 deg for the first three hangers to almost zero degrees for the hangers in mid-span. The hangers are locked-coil strands of 70 mm diameter (Fig. 3) with high-density polyethene (HDPE) sheathing, bringing the outer diameter to 84 mm with a linear mass density of 30 kg/m (Jamal and Sundet, 2021). The thick HDPE sheathing, instead of the traditional painting, was chosen in order to protect the hangers from the environment. However, this increased the diameter of the hangers D with only a negligible increase in mass per unit length m. As a consequence, this considerably reduced the ratio between the structural damping and the aerodynamic excitation, represented by the Scruton number in Eq. (2). With the sheathing, Sc is reduced by approximately 30% (from 15 to about 10.5, if the structural damping decrement of $\delta_s = 0.0015$ is considered, measured by logarithmic decrement in Larsen et al. (2022)). This change made the hangers particularly susceptible to vortex-induced vibration. Previous studies on the Hardanger bridge, which has the same locked coil cable hangers but without HDPE sheathing (Cantero et al., 2018), showed the complexity of an accurate estimation of the structural damping for each mode of vibration. However, it is shown that the hanger damping is in general very low ($0.001 < \delta_s < 0.0015$) and tends to decrease slightly for higher modes.

2. Monitoring of a long-span suspension bridge

2.1. A long-term monitoring project

During the construction of the Hålogaland Bridge, the longest hangers experienced strong vibrations, which continued after the project's completion. Concerns regarding the hangers' long-term durability led to the implementation of dampers on all hangers as a preventive measure. It was decided to use Stockbridge dampers (Fig. 5); this solution proved to work, significantly reducing the amplitude of vibration. However, shortly after installation, half of the installed dampers were damaged. The causes of the failure of the devices are still unclear since no structure monitoring was available at the time. However, it has been observed that the component that failed was the messenger cable near the clamp between the damper and the hanger. Pictures of some of the failed dampers are reported in Fig. 4. A similar failure mechanism has already been observed on the HPDE-coated hangers of the Çanakkale bridge in Türkiye (Larsen, 2024).

Following this event, the whole set of about a hundred dampers has been replaced with a modified improved version provided by the sup-

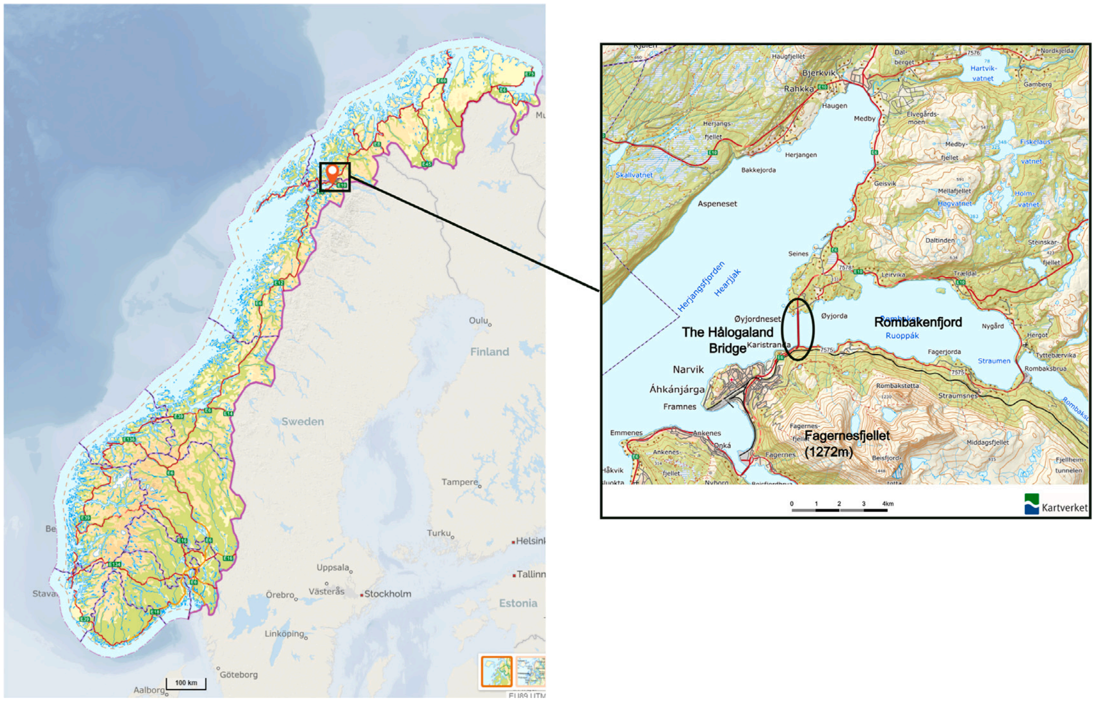
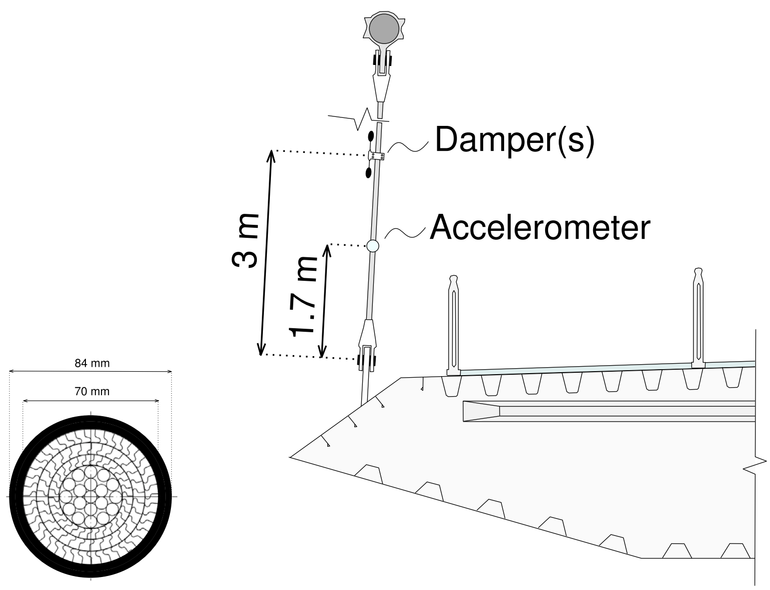


Fig. 2. Location and surrounding topography (map images from Kartverket©).



(a) (b)

Fig. 3. (a) Dimensions of the locked-coil hanger with sheeting and (b) placement of accelerometers and damper.

plier (Fig. 6(a)). Furthermore, a collaboration between the Norwegian Public Roads Administration (NPRA) and the Norwegian University of Science and Technology (NTNU) allowed the start of an extensive long-term monitoring project on cable vibration and global dynamics of the structure. The measurement campaign was started in May 2021 and continued with minor interruptions until the present time. The whole bridge was provided with a comprehensive measurement system described in detail in Petersen et al. (2021). The present work, however, focuses on data from three accelerometers placed on the first three hangers on the South side of the bridge. The wind data is taken from the anemometer nearest to the considered hangers, as shown in the scheme in Fig. 7.

2.2. Monitoring system and data collection

The three triaxial accelerometers (Dytran 3063B), installed on the three longest hangers, were located 1.70 m from the lower end of each hanger. The anemometer on hanger #7, as shown in the scheme in Fig. 7, was placed at a height of 8 m above the bridge deck. Although this anemometer is located 80 m from the instrumented hangers, it

(a) (b) (c)

Fig. 4. Pictures of the failed dampers ©Statens Vegvesen. (a) Failed damper on hanger #48. (b) Section of failed messenger cable on damper #50. (c) Damper with messenger cable partially failed.

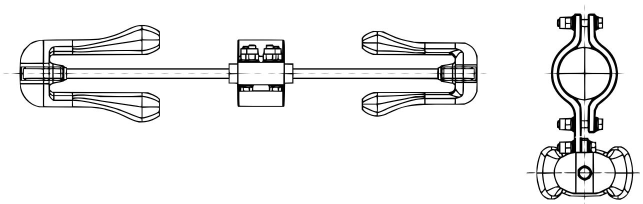


Fig. 5. Drawing of the Stockbridge damper designed for the Hålogaland bridge (Larsen et al., 2022).

is assumed that the important statistical metrics such as mean wind speeds and turbulence intensities, which will be discussed later, are representative of the locations of the hangers. This assumption is supported by the fact that in the context of wind engineering, the analysis of VIV places considerable emphasis on the system's slow dynamics (Denoël, 2020), primarily influenced by the slowly fluctuating mean wind speed. Consequently, the slow dynamic characteristics of the wind encountered by the hangers will strongly correlate with those measured by the anemometer situated 80 m away.

The accelerations are sampled at 200 Hz, while the anemometer data has a sampling frequency of 32 Hz. The initial Stockbridge dampers layout consisted of two dampers on each hanger, one aligned with the bridge girder and one aligned transverse to the bridge girder. However, after a storm in 2021, the first hanger was left with no dampers, the second hanger with only one in the transverse direction, and hanger #3 with both dampers undamaged. Data recording started on the 25th of May 2021, shortly after most of the dampers failed. This allowed the collection of acceleration data for hanger #1 without any damper, highlighting the entity of the VIV issue and allowing for direct evaluation of the performances of two different damper configurations in the two neighbouring hangers.

In December 2021, all the dampers were replaced. The geometric characteristics of the new dampers are summarized in Table 1. The first three longest hangers, on both sides of the bridge, have been equipped with three dampers: one in the cross-bridge direction and two in the along-bridge direction. Therefore, in all the measurements following the 1st of February 2022, the instrumented hangers shown in Fig. 7 are equipped with three Stockbridge dampers. As shown in Fig. 8, all three are clamped in the same location 3 m above the deck. As will be discussed in Section 3.2, this distribution might be suboptimal, since dampers' dissipative influence on the vibration modes is concentrated in one point.

The data considered in this paper extends until June 2023, with two main downtimes during the fall of 2021 and the summer of 2022, making a total of 15176 hours of available measurements.

3. Analysis of the data with standard statistical processing tools

3.1. Wind field description

Wind velocities in the present study were recorded in polar coordinates with the anemometer described in Section 2. From the measured wind data, the 10-minute mean wind magnitude $\bar{U}$ and direction $\bar{\theta}$ were determined. Next, the instantaneous wind speed has been decomposed in the alongwind and crosswind components:

$$U_u(t) = U(t)\cos(\Delta\theta(t)) \tag{3a}$$

$$U_v(t) = U(t)\sin(\Delta\theta(t)) \tag{3b}$$

Here, $U(t)$ is the instantaneous wind magnitude and $\Delta\theta(t)$ is the difference between the instantaneous direction $\theta(t)$ and the average direction $\bar{\theta}$ for the 10-minute time interval. The 10-minute averaged wind velocities for all recordings from the anemometer on hanger #7 are shown in the wind rose in Fig. 9(a). From this density plot and the histogram in Fig. 9(b), it can be seen how the majority of the wind data has a mean wind speed of about 6–7 m/s and has either East or West as the main direction. As expected, the topography on the site makes it unlikely that winds are approaching from the North or South, where larger landmasses and mountains are located. The Westerly winds are characterized by a wider directional angle compared to the Easterly winds. The reason for this can be found in the topography surrounding the bridge site, as shown in Fig. 2. The Easterly wind is bounded by the narrow Rombaken fjord with steep mountainsides, while to the West side is a wider fjord basin which is less influenced by high mountains. From both directions, the highest 10-minute averaged wind speeds were approximately 22 m/s. The turbulence intensities in the alongwind and crosswind directions are defined as in Eq. (4), where σ_u and σ_v are the 10-minute standard deviations of the wind components U_u and U_v in Eq. (3).

$$I_u = \frac{\sigma_u}{\bar{U}}, \quad I_v = \frac{\sigma_v}{\bar{U}} \tag{4}$$

(a) (b)

Fig. 6. Sensors on the hangers: (a) triaxial piezoelectric accelerometer and two of the new dampers. (b) WindMaster 3D Pro fixed on hanger #7.

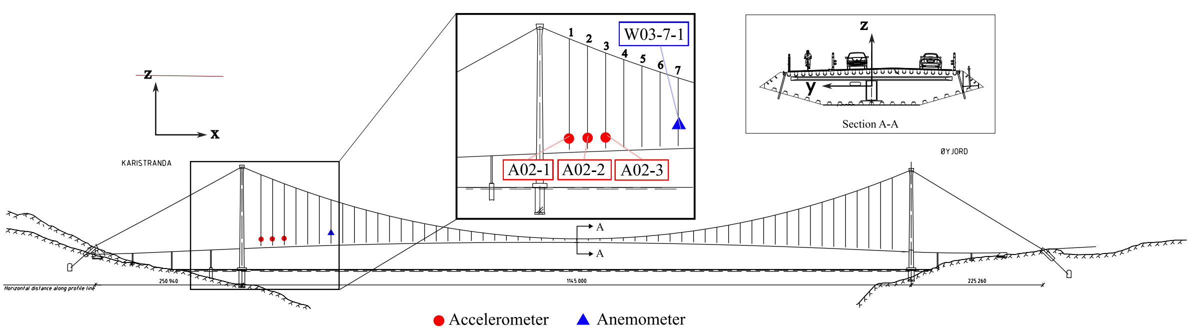


Fig. 7. Scheme of the positions of the sensors on the Hålogaland bridge that are used in this work. The red circles show the accelerometers on the three longest hangers near the South tower. The blue triangles show the anemometer placed on hanger #7.

Table 1
Geometric parameters of the Hålogaland Stockbridge dampers.

Upper mass [kg]	Lower mass [kg]	Length messenger cable [mm]	Diameter messenger cable [mm]
5.1	8.5	250	16

Fig. 10 shows the distribution of turbulence intensities for the whole dataset. The two distributions are very similar, with a Pearson correlation coefficient of 0.83. Therefore, in the following figures, only the crosswind turbulence intensity will be shown. Figs. 11(a) and 11(b) show the turbulence intensity as a function of wind direction and wind magnitude, respectively. In Fig. 11(b), it can be seen how, for high wind velocities, the turbulence intensities converge to 10%. Most of the data points with an average wind speed greater than 5 m/s are characterized by turbulence intensities lower than 30%.

3.2. Dynamic response of the hangers

In the data from the triaxial accelerometers, only the acceleration in the horizontal plane has been considered (plane $x - y$ according to the fixed reference system shown in Fig. 7). The acceleration has been transformed into a non-fixed reference system aligned with the 10-minute average wind direction denoted as alongwind and crosswind acceleration. The subsequent analysis in this work will concentrate on crosswind acceleration and displacement, as these are the primary components of vibration significantly influenced by VIV. The spectrum analysis of the three hangers' responses facilitated the calculation of their fundamental frequencies, denoted as f_i. These frequencies were determined by the difference between two adjacent frequencies in the spectrum. The fundamental frequencies and other essential geometrical characteristics of the hangers are summarized in Table 2. According to taut string theory, the speed of bending waves propagating along the hanger is described by:

$$c = 2 f_i L \tag{5}$$

where f_i is the fundamental frequency and L is the length of the hanger. Approximating the hanger as a taut string permits the estimation of the tension in the hangers due to dead load, expressed as

Table 2
Properties of hangers number 1, 2 and 3.

	Diameter [m]	Tension [kN]	Linear mass density [kg/m]	Length [m]	Inclination [deg]	Fundamental frequency f_i [Hz]
Hanger #1	0.084	1045	30	102.2	2.8	0.78
Hanger #2	0.084	966	30	111.5	2.8	0.80
Hanger #3	0.084	911	30	103.2	2.8	0.84

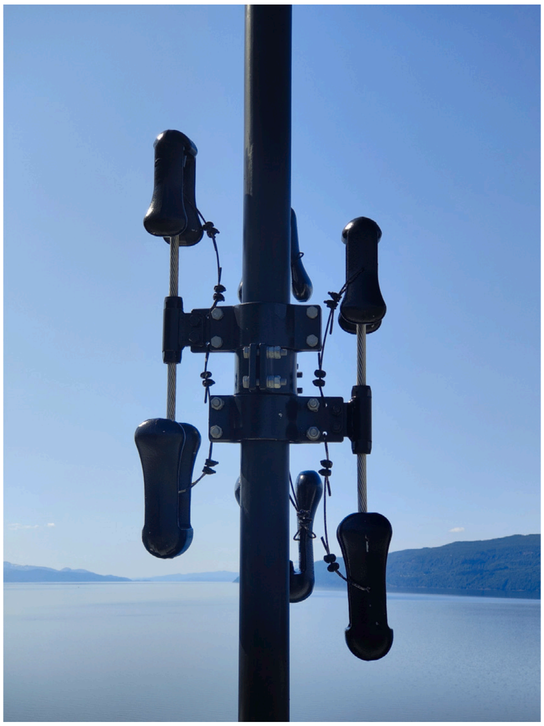


Fig. 8. New configuration of dampers for the first three hangers from February 2022.

$T = mc^2$. Assuming hinged boundary conditions, the mode shapes are given by:

$$\phi_n(x) = \sin\left(\frac{\pi}{\lambda_n}x\right), \quad \lambda_n = L/n \tag{6}$$

Fig. 12 illustrates various vibration modes for the three hangers, highlighting the locations of the accelerometers and the dampers. The figure indicates that while the dampers effectively target certain modes, their influence is considerably less on others, given that there are potentially tenfold VIV modes. Importantly, the modal influence on the acceleration sensor varies between the modes, which in turn also affects the calculated response statistics.

In Figs. 13 and 14, two long time histories of the crosswind acceleration response of hanger #1 are shown. These time histories have been selected for the almost constant slowly varying incoming wind. Figs. 13(a) and 14(a) show also the Short-time Fourier transform of the acceleration calculated by splitting the data into segments of length 256 points (corresponding to 2 s) with an overlap of 128 points. For each segment, the spectrum is calculated with the FFT algorithm and a Hanning window is applied to reduce end effects. Fig. 13 shows how the condition of almost constant slowly varying wind magnitude is a necessary but not sufficient condition to have a single lock-in vortex-induced vibration. Other factors govern the entrance and exit from a specific lock-in condition of the hanger, like the coherence of the wind velocity along the hanger and the change in wind direction. However, inferring further conclusions without multiple measurement points for the wind profile along the hanger is also difficult. Additionally, it can be seen that the vortex-induced response takes several minutes to reach a steady state.

From the spectrum depicted in Fig. 13(b), corresponding to the acceleration segment centred in 292 s, the vibration is mainly monoharmonic at around 37 Hz, corresponding to mode 48. The observed frequency of vibration aligns reasonably well with the Strouhal frequency of vortex shedding, f_{vs} (Eq. (1)), indicated by the dashed line in the graph. However, a gap of approximately 5.5 Hz between the two frequencies is observed. Inverting Eq. (1), this gap corresponds to a wind speed difference of about 2.57 m/s. This variation could potentially be attributed to turbulence fluctuations occurring between the hanger and the location of the anemometer. In the second time instant reported in Fig. 13(c) instead, a vibration with a wider frequency content can be observed. Mode 48 is still present, but with a contribution from mode 43 (33.4 Hz) and other lower modes.

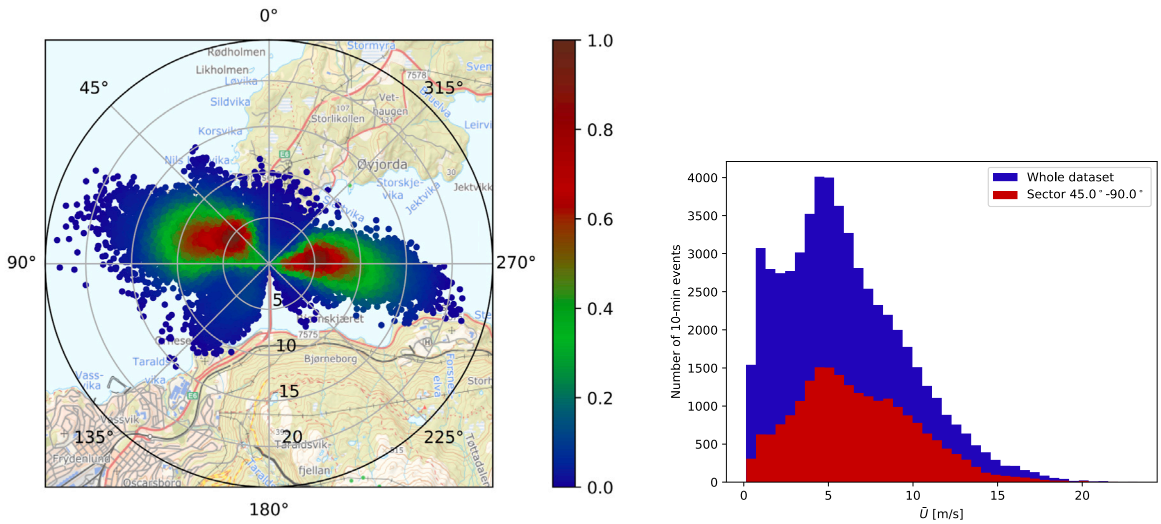
A second interesting example recorded the same day is shown in Fig. 14. From Fig. 14(b) it can be seen that the main frequency content is at about 24 Hz corresponding to mode 31, but there is also a clear component at about 20 Hz corresponding to mode 26. In this case, the vibration frequency is nearer to the theoretical Strouhal frequency for the instantaneous wind speed. In both Figs. 14(b) and 14(c), a spectral content is also visible at about the double of the crosswind frequency of vibration: at about 44 Hz for Fig. 14(b) and about 48 Hz for Fig. 14(c). This most likely corresponds to the inline frequency of vibration, $f_{inline} = 2f_{VIV}$. Analysis of Figs. 14(b) and 14(c) initially suggests that the hanger is experiencing multimode vibration. Yet, the underlying physical phenomenon appears to be more interesting. A closer examination, shown in Fig. 15, reveals that the hanger's behaviour is not a continuous vibration at the spectrum indicated in Fig. 14(c). Instead, the hanger displays a beating behaviour. This pattern involves groups of vibrations, each consisting of approximately 20 cycles at a consistent frequency. After this period, the vibrations diminish almost to zero. This process is then repeated with a frequency in the order of 1 Hz. This behaviour could be explained by the phenomenon of beating, which occurs when two oscillators of very similar frequencies interact. This interaction results in a constructive and destructive interference, modulating the amplitude of the response at a frequency equal to the difference between the interacting frequencies. Triggering responses in several modes can be caused by vortex waves propagating along the hanger, from regions of high wind velocity to those of lower velocity, that can create moving cells with slightly different vortex detachment frequency (Oh et al., 2022). This can result in the observed beating behaviour. This is also facilitated by the very low structural damping of the hanger and the very closely spaced natural frequencies. This behaviour is frequently observed in the data set, often between two lock-in monoharmonic vibration episodes.

3.3. Statistics of hangers' VIV

Statistics have been derived by processing the acceleration data. In particular, the maximum a_{max} and the standard deviation σ_a of the acceleration response metrics of interest. The accelerations are also integrated in the frequency domain to obtain displacement time histories:

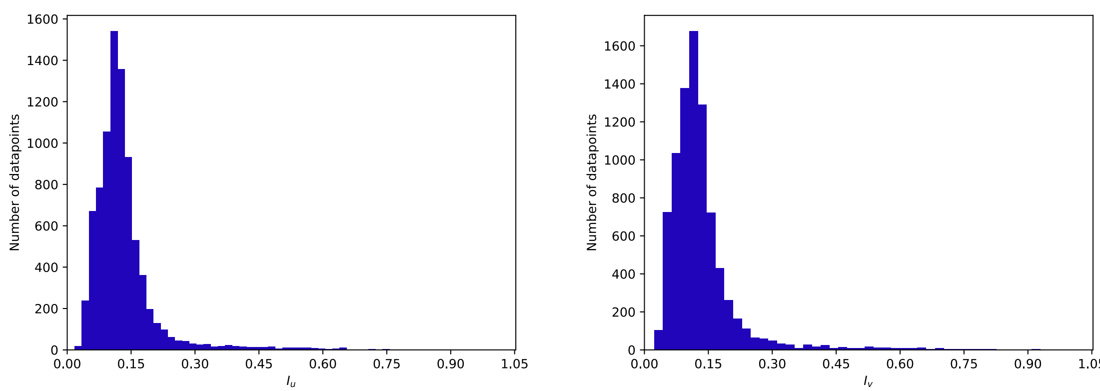
$$Y(\omega) = \frac{\ddot{Y}(\omega)}{-\omega^2} \tag{7}$$

Here, $Y(\omega), \ddot{Y}(\omega)$ are the Fourier transform of the displacement and the acceleration, respectively. The Fourier transform of the acceleration $\ddot{Y}(\omega)$ is filtered with a rectangular filter with cutoff frequency 2 Hz.



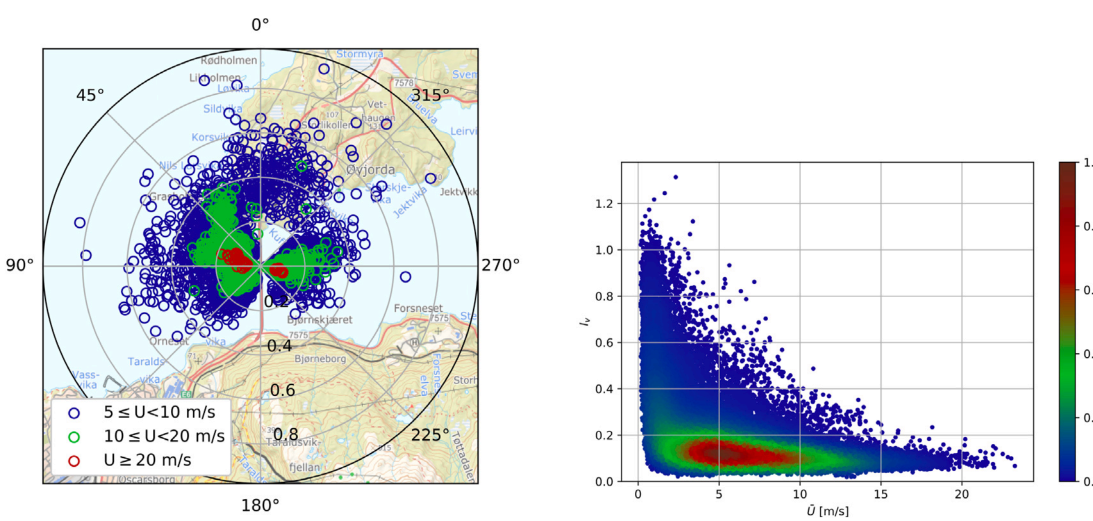
(a) (b)

Fig. 9. (a) Wind-rose plot of 10-min averaged wind velocity (the colour bar indicates the relative density of the data points) and (b) corresponding histogram, for the whole data set and the sector 45°–90°, used in Section 4.2.



(a) (b)

Fig. 10. Histogram of alongwind turbulence intensity (a) and crosswind turbulence intensity (b) of the entire data set.



(a) (b)

Fig. 11. (a) Wind-rose plot of crosswind turbulence intensity I_v. (b) Crosswind I_v turbulence intensity versus 10-minute mean wind speed (the colour bar indicates the relative density of the data points).

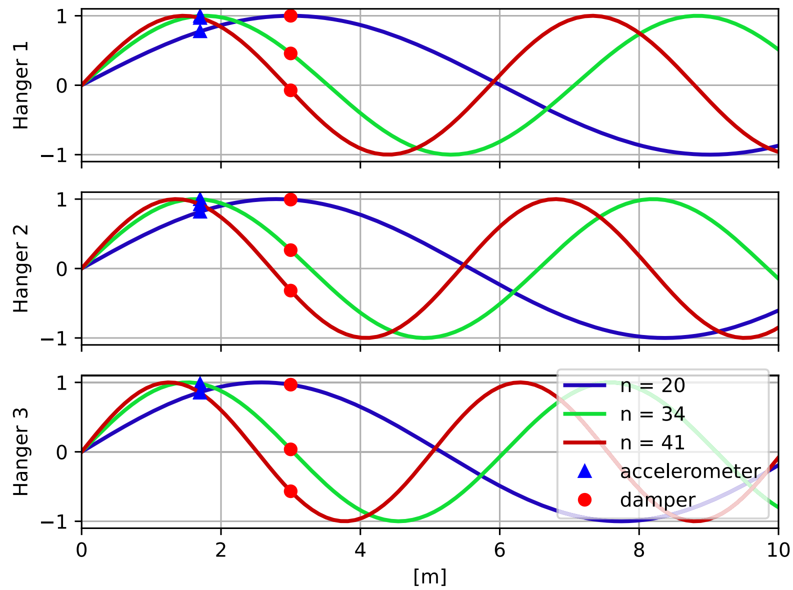


Fig. 12. Examples of mode shapes of the three hangers according to taut string theory. The blue triangle and the red circle indicate, respectively, the position of the accelerometer and the dampers on each of the hangers. Here are plotted the first 10 m of the hangers. The natural frequencies are integer multiples of the fundamental frequency reported in Table 2.

Then, the displacement $y(t)$ is found by taking the inverse Fourier transform of $Y(\omega)$. For displacement, the same statistical indicators of maximum and standard deviation are calculated over intervals of 10 min. As explained in Section 2.2, the two years of recordings allowed a critical comparison of the effect of different damper configurations on the hangers. Fig. 16 shows the standard deviation of the displacement of hanger #1 from 2021, when it had no dampers, to the following period when three dampers were installed, as a function of the 10-minute averaged wind speed. The introduction of three dampers into the system resulted in the reduction of the maximum displacement of about 80%. However, it is interesting to notice how the frequency band in which the dampers are effective is limited to the range 4 m/s and 15 m/s. At wind speeds below 4 m/s, the displacement of the hanger is unchanged. The comparison between the configuration with one and three dampers can be seen on hanger #2 in Fig. 17(a). The additional dampers slightly improve the reduction of the displacement between 10 and 15 m/s. However, no improvement is seen for the low frequencies. Currently, all the dampers have been clamped 3 m from the lower hanger's anchorage (see Fig. 8). The authors aim to investigate if better performance can be achieved by optimizing the locations of the dampers (Di et al., 2020).

The scatter plots in Figs. 16(a) and 17(a) have been re-plotted in Fig. 18 transferring the displacement to the value at the antinodes of the vibration modes. Assuming the mode shape as represented in Eq. (6), the modal displacement, for mode n, at the sensor position x_s can be written as in Eq. (8). The value of $n = f/f_i$ can be found assuming the hangers fundamental frequencies reported in Table 2.

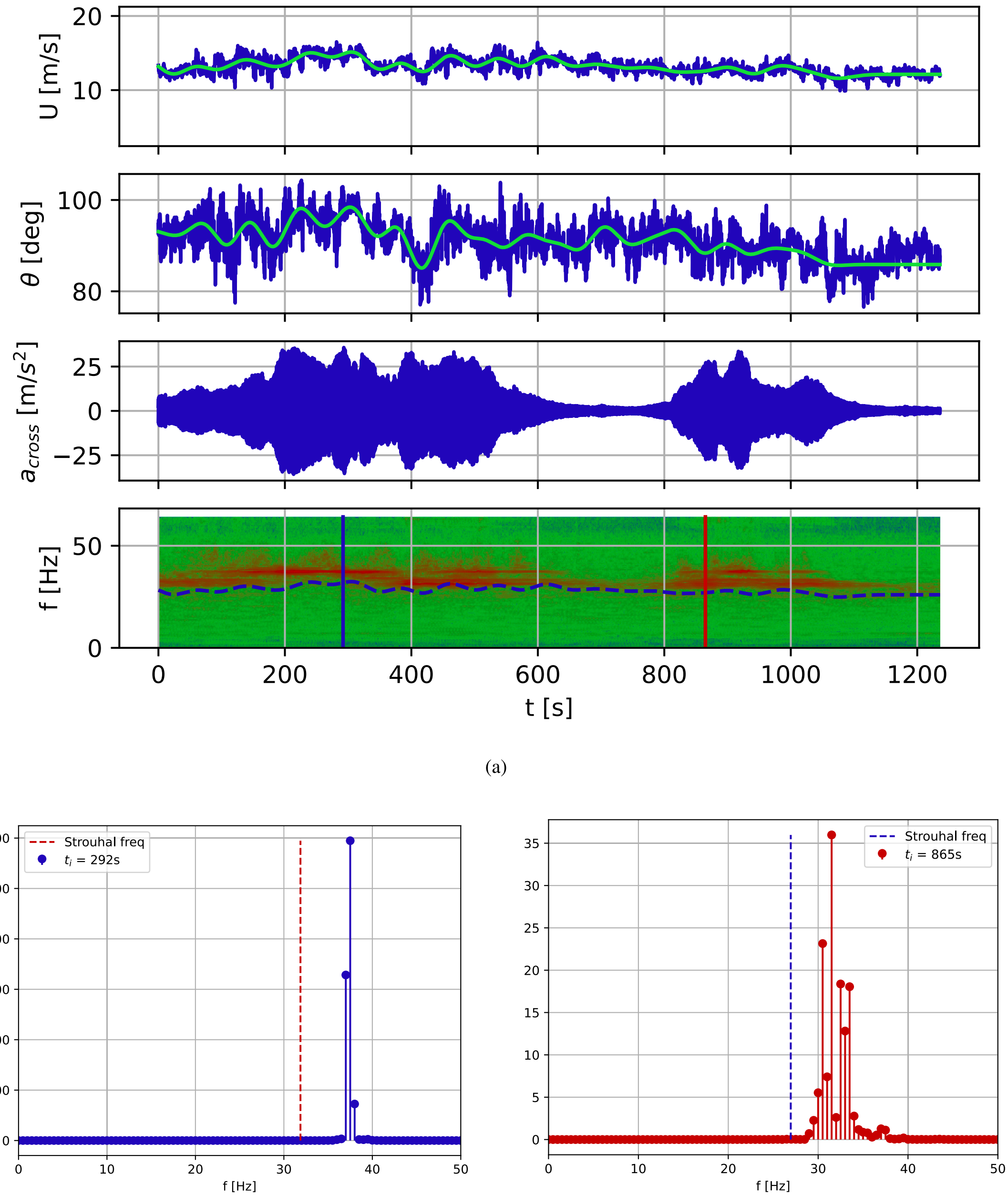
$$y(x_s) = A_n \Phi_n(x_s) = A_n \sin\left(\frac{\pi}{\lambda_n} x_s\right) \tag{8}$$

Therefore, the amplitude at the antinode A_n, is found dividing the measured displacement by the instantaneous value of the mode shape at the sensor location $\Phi_n(x_s)$, assuming monoharmonic vibration and fixed Strouhal number. From this rescaling of the data, aiming at showing statistics for modal amplitudes rather than displacements at a fixed location, it can be noticed how the highest displacements happen for low wind speeds, corresponding to the frequency range where the dampers are the least effective. However, since the mode shape and fundamental frequencies are only known approximately, this representation must be regarded as an approximation. Given this uncertainty, combined with the challenges posed by the very closely spaced natural frequencies of the hanger, the subsequent analysis in the paper will rely on the displacement measured by the accelerometer.

Finally, in Fig. 19 the statistics from all four encountered damper configurations are plotted together, both in terms of a scatter plot of the standard deviation of displacement (Fig. 19(a)) and in terms of a cumulative distribution of the maximum displacement (Fig. 19(b)) at the accelerometer. The difference in damping effectiveness from the undamped hanger to the hanger with three dampers can easily be observed. It must be highlighted that, from Fig. 19(a), the solution with three dampers on hanger #1 seems less effective than the solution with two dampers on hanger #3. This is only caused by the fact that the accelerometers are all placed at the same height of 1.7 m, regardless of the hanger. For example, given a wind speed of 15 m/s, the 41st mode is excited on hanger #1. For this mode the accelerometer position corresponds to 97% of the maximum amplitude point, see Fig. 12. While on the slightly shorter hanger #3, the 38th mode is excited. For this hanger, the accelerometer position corresponds to only 92% of the point of maximum amplitude, resulting in a smaller perceived displacement through the measurements.

The effect of turbulence intensity on the response can be appreciated from Fig. 20, where the influence of both wind magnitude and turbulence intensity is emphasized. At around 7 m/s, as mentioned earlier, hanger #1 without any damper shows the maximum VIV amplitude, but this seems only observable for low turbulence intensities. This is expected since, with little turbulence, a coherent vortex detachment along the hanger can be generated more easily, and therefore, a precise frequency is excited for a longer time, enabling it to enter the lock-in limit cycle with the highest displacement. When turbulence is high, the coherent vortex-shedding is continuously disturbed, resulting in a lower maximum displacement (Denoël, 2020).

The statistical indicators shown in Figs. 16 to 20 were calculated over for 10-minute intervals. Fig. 21 illustrates an example of two intervals, showing the wind magnitude and direction and the acceleration response of the three hangers. The spectrogram of hanger #1 is also shown. The spectrograms in Fig. 21 show how the frequency content of the hanger vibration closely follows the frequency of vortex-shedding f_{vs} (Eq. (1)), which is indicated with a continuous blue line. The alignment of the vibration frequency with the vortex-shedding frequency indicates that the resonance excitation witnessed can indeed be attributed to vortex-induced vibration. As shown in Section 3.2, the



(a)

(b)

(c)

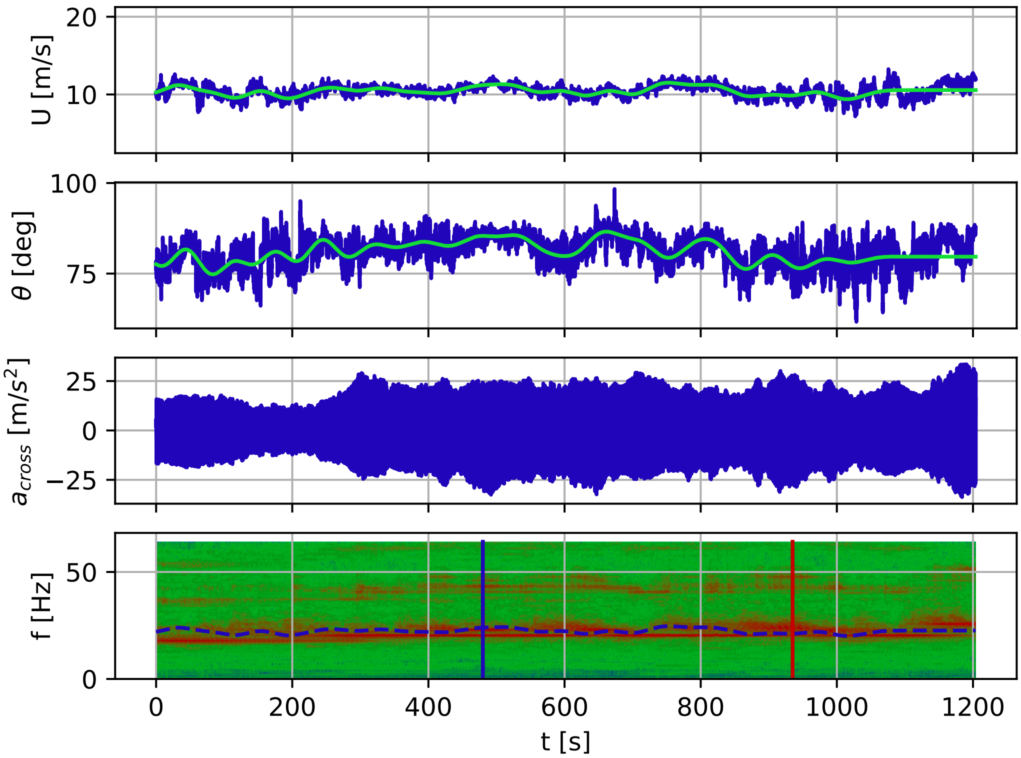
Fig. 13. Sample time history recorded on hanger #1, without any damper, on 19/7/2021. (a) Time history of wind magnitude U, wind direction θ from the anemometer and crosswind acceleration from the accelerometer on hanger #1 and its spectrogram. Vertical blue and red lines indicate the time instant of the spectra plotted in Figs 13(b) and 13(c) respectively. (b) Spectrum of the acceleration for the segment centred at 200 s from the beginning of the window. (c) Spectrum of the acceleration for the segment centred at 921 s from the beginning of the window.

time required to reach a steady state for the vortex-induced response is in the order of 10 min. Therefore, it could easily happen, as shown in Fig. 21(b), that only part of the process is included in the time interval used to calculate the statistics and the VIV episode develops across multiple intervals. Vice versa, multiple VIV episodes can be captured in the same observation interval, as shown in Fig. 21(a). The consequence of this is seen in Fig. 20(a), where a scatter of numerous oscillation amplitudes is observed for the same averaged wind speed.

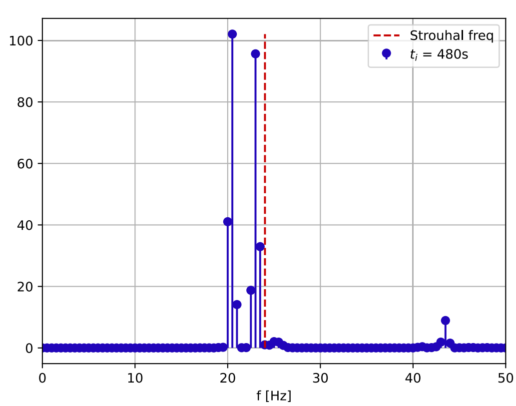
4. Advanced statistical methods for the analysis of time series

As highlighted in Section 3, using a fixed time interval of 10 min in the statistical analysis might lead to the recording of multiple growth-to-steady state VIV episodes or suffer from episodes spreading on consecutive windows. This causes, as shown in Fig. 20(a), a scatter of different oscillation amplitudes for the same wind speed. If standard deterministic models, like the one proposed by Fazl Ehsan and Scanlan (1990), were used to predict this type of response, one would expect it to cluster along the upper envelope in this diagram. The scatter is not only attributable to variability in the measurements (including the turbulence and uncertainty on the wind field) but also to the possible non-stationary nature of the acceleration records over the considered time windows of 10 min duration.

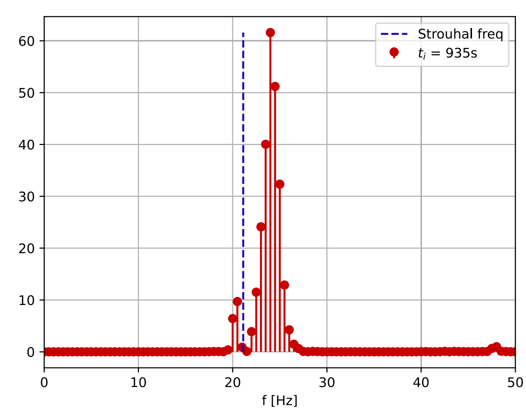
To investigate possible causes of this scatter, the wind data was analysed by searching for time intervals with only a small slowly varying wind variation. The method used to define almost constant slowly varying wind is described later in Section 4.1. As will be shown, for the majority of the time, the slowly varying wind is almost constant



(a)



(b)



(c)

Fig. 14. Sample time history recorded on hanger #1, without any damper, on 19/7/2021. (a) Time history of wind magnitude U, wind direction θ from the anemometer and crosswind acceleration from the accelerometer on hanger #1 and its spectrogram. Vertical blue and red lines indicate the time instant of the spectra plotted in Figs. 14(b) and 14(c) respectively. (b) Spectrum of the acceleration at 480s from the beginning of the window. (c) Spectrum of the acceleration at 935s from the beginning of the window.

for 3 min or less. The fact that the slowly changing mean wind speed changes considerably within these 10-minute intervals, is critical for VIV mechanics, as they typically require a relatively steady wind velocity to develop lock-in vibrations fully. This naturally calls for a study of the influence of the time window T used to perform the statistical analysis of VIV data. This analysis is exposed in Section 4.2.

4.1. Wind data analysis

The wind data, which is continuously recorded (except for occasional system downtime), has been sliced into time intervals with almost constant slowly varying wind speeds. In this work, a time interval with almost constant slowly varying wind speed is defined as a portion of time history where the minimum u_{min} and maximum u_{max} slowly varying wind speed lies within ±15% of the average wind speed of the window, $\bar{U}$. The slowly varying wind speed is obtained by low-pass filtering the instantaneous wind speed using a 20th order Butterworth filter with a cut-off frequency of 1/60 Hz. An algorithm to identify time intervals with almost constant slowly varying wind has been developed and is presented in Algorithm 1. The algorithm initially considers only the first sample of the wind velocity. Next, samples are added individually as long as the slowly varying wind speed at the particular time instance is within the boundaries explained above. The algorithm stops and defines the end of the time interval when the criteria are not met. An example is shown in Fig. 22, illustrating that whenever the filtered wind speed exceeds the boundaries defined by

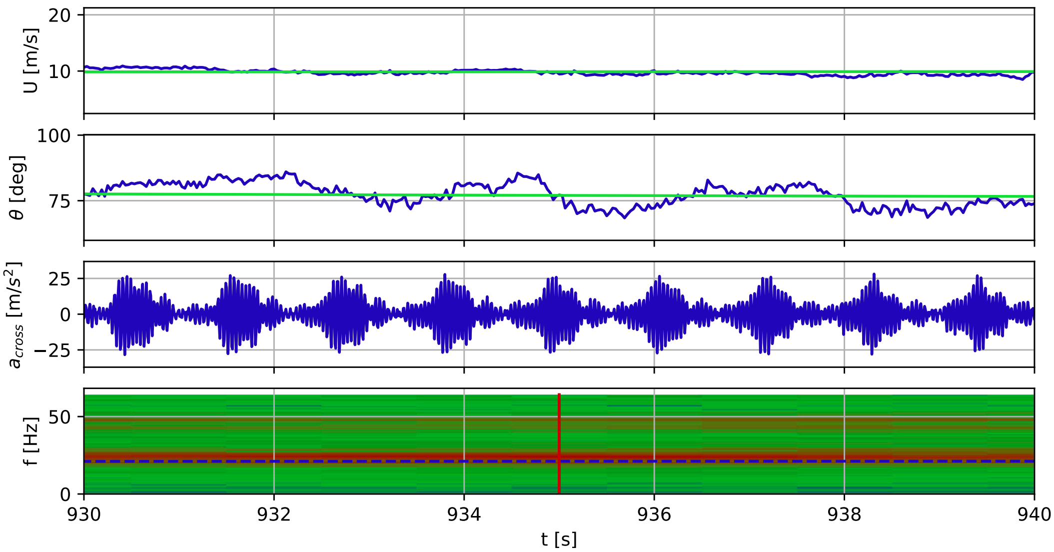
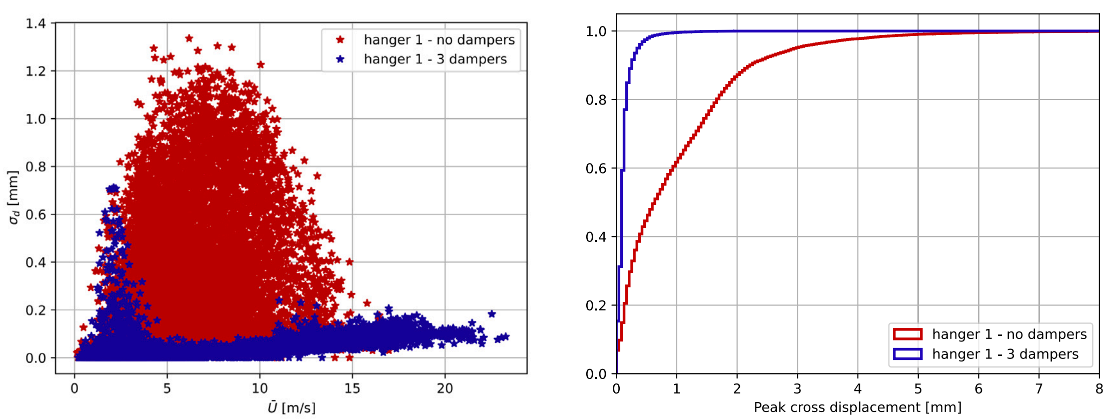
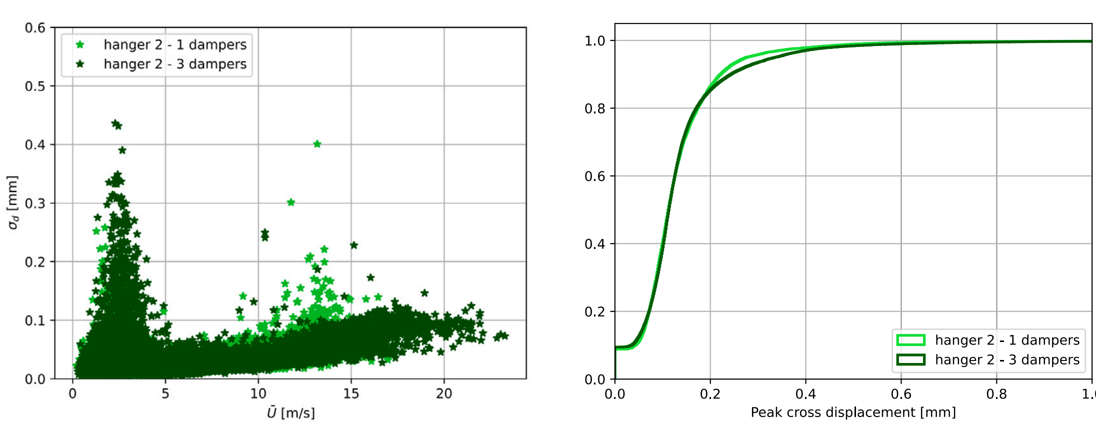


Fig. 15. Portion of 10 s from the time history in Fig. 14(a), corresponding to the spectrum shown in Fig. 14(c).



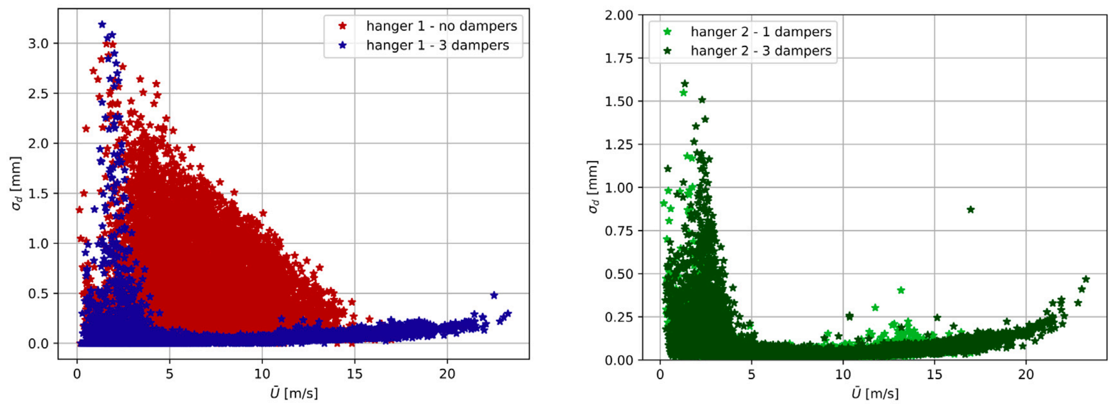
(a) (b)

Fig. 16. Comparison of performances of two-damper configurations on hanger #1 through displacement at 1.7 m from hanger's anchorage. (a) Scatter plot of the standard deviation of the displacement versus 10-minute mean wind velocity, (b) cumulative histogram of the peak displacement.



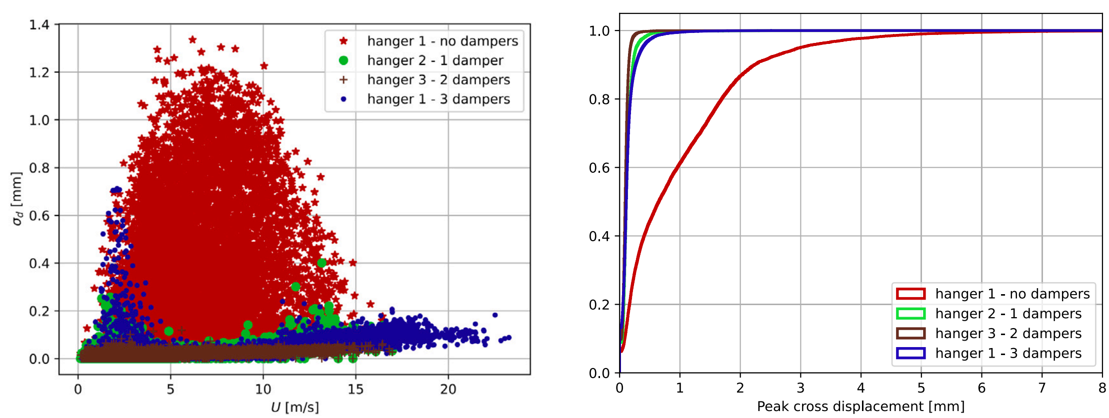
(a) (b)

Fig. 17. Comparison of performances of two-damper configurations on hanger #2 using displacement at 1.7 m from hanger's anchorage. (a) Scatter plot of the standard deviation of the displacement versus 10-minute mean wind velocity, (b) cumulative histogram of the peak displacement.



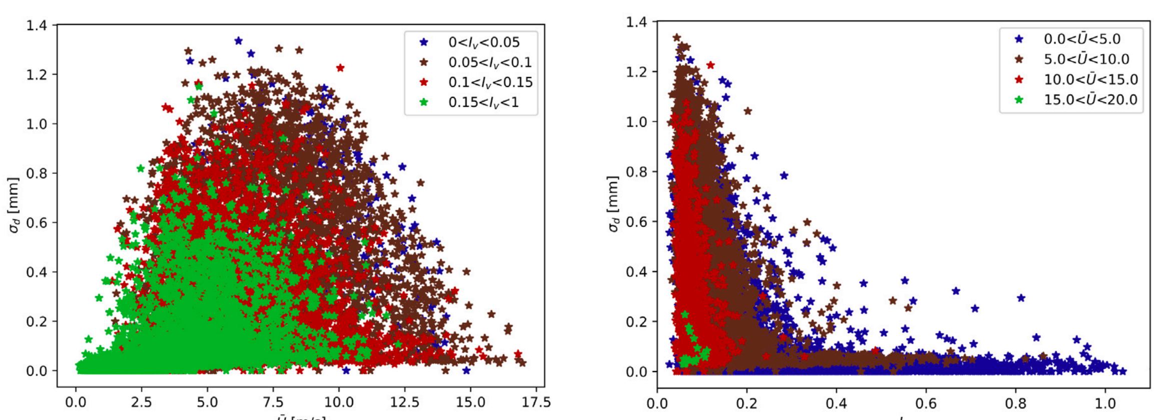
(a) (b)

Fig. 18. Standard deviation of the displacement transferred to the value at the antinodes. (a) Scatter plot for hanger #1 versus 10-minute mean wind velocity, (b) Scatter plot for hanger #2 versus 10-minute mean wind velocity.



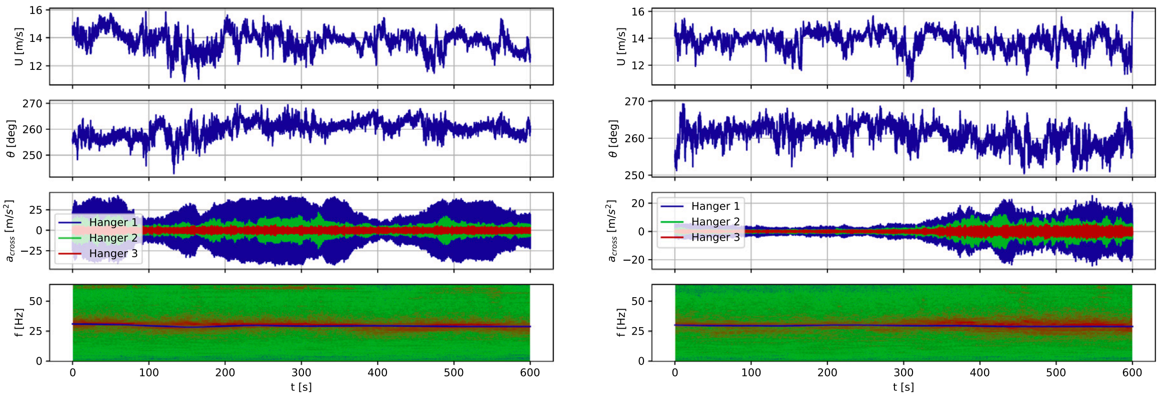
(a) (b)

Fig. 19. Comparison of performances of four different damper configurations on the three analysed hangers through displacement at the accelerometer. (a) Scatter plot of the standard deviation of the displacement versus mean wind velocity, (b) cumulative histogram of the peak displacement.



(a) (b)

Fig. 20. Standard deviation of the displacement on hanger #1 during 2021 (no damper) as function of (a) wind speed and (b) crosswind turbulence intensity. The statistics are calculated for $T = 10$ min.



(a) Recording started 18:49.

(b) Recording started 18:38.

Fig. 21. Sample time histories from 15th of June 2021. In blue, from the top, the wind magnitude, the wind direction and the crosswind acceleration of hanger #1, with no damper, hanger #2, with one damper, and hanger #3, with two dampers. In the bottom plot is shown the spectrogram of the acceleration of hanger #1. The blue line in the spectrogram shows the theoretical vortex-shedding frequency corresponding to the average wind velocity.

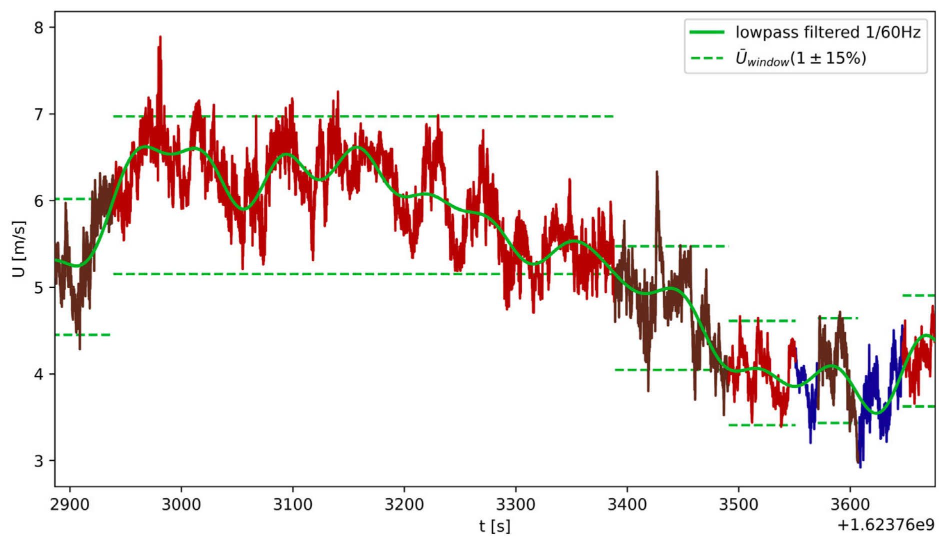


Fig. 22. Example of the application of the algorithm for identification of almost constant slowly varying wind windows. The results show how both long and short windows are identified. The windows highlighted in blue are discarded as their average wind speed is below 4 m/s.

the green dashed lines, the time interval ends, and a new one is started. Time intervals with average wind less than 4 m/s have been discarded.

Algorithm 1 Inclusion of a new data point u into an almost constant slowly varying wind window

1: Initialize the window's current average $\bar{U}_{old}$, number of samples n_{old}, maximum u_{max} and minimum u_{min}.
2: Update $\bar{U}_{new} = \frac{\bar{U}_{old} n_{old} + u}{n_{old}+1}$ with the new point u
3: Update window extremes u_{max} and u_{min} with the new point u, if required.
4: **if** $(u_{max} < 1.15\bar{U}_{new})$ & $(u_{min} > 0.85\bar{U}_{new})$ **then**
5: u can be added to the window
6: $\bar{U}_{old} = \bar{U}_{new}$
7: $n_{old} = n_{old} + 1$
8: Go to 1.
9: **else**
10: Discard u and start a new window.
11: **end if**

Fig. 23 shows the outcome of the described analysis in terms of the duration of time intervals with almost constant, slowly varying wind speeds. The plot shows that 50% of the time intervals are shorter than 3 min. Fig. 21(b) showed that it could also take 5 min for VIV to reach the limit state. This means that, in many cases, there is not enough time for the hanger to reach a steady state before the wind speed changes. From these results, it can be seen why the long-term distribution of the response is highly skewed towards small amplitudes, with occasionally larger peaks when the lock-in limit cycle is reached, as depicted in Fig. 20(a). However, the plot in Fig. 23 also shows that there are rare instances of time intervals with almost constant, slowly varying wind that can last for 30 min or more.

The definition of time interval with almost constant slowly varying wind speed was formulated only concerning the magnitude of the wind vector. This choice was made for two main reasons. First, the overall wind distribution indicates a prevalence of Easterly and Westerly winds, resulting in mainly two perpendicular directions of wind incidence to the bridge. Second, given the axisymmetric geometry of the considered system, minor changes in the incidence angle were assumed to be negligible. However, it is probable that including the direction in the window definition will result in a further reduction of the average time interval length, a hypothesis which has not yet been verified.

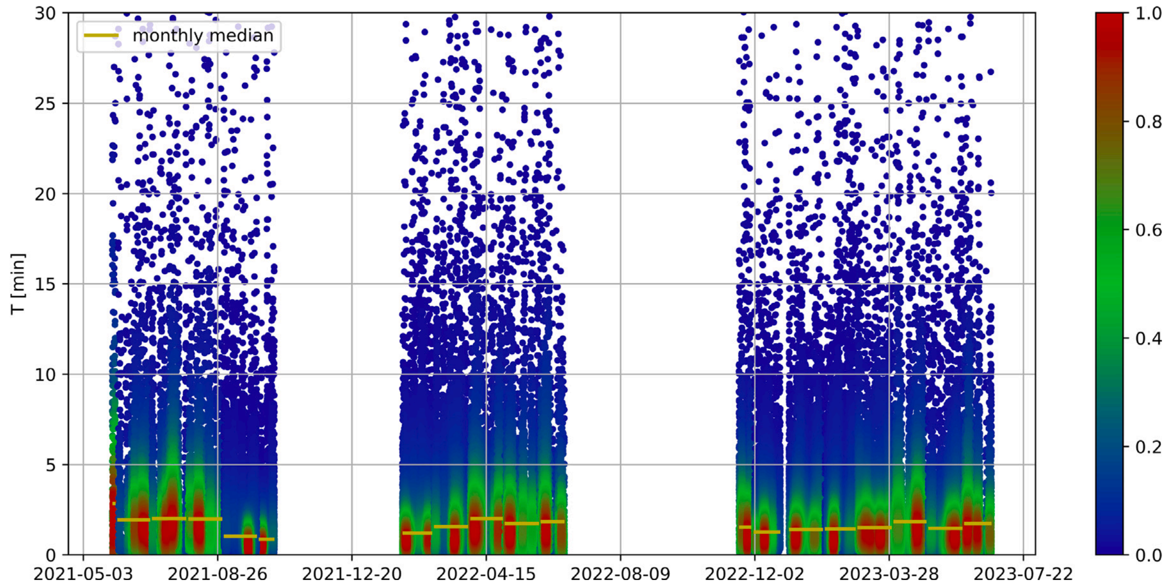


Fig. 23. Length of time intervals with almost constant slowly varying wind for the whole data set. In yellow is the median value for each month. The colour bar indicates the data density.

4.2. Response analysis

The mean, maximum, and standard deviation, serving as statistical indicators of the response, are calculated across six distinct time intervals $T = \{30\text{ s}, 60\text{ s}, 5\text{ min}, 10\text{ min}\}$. This analysis seeks to determine if the statistical properties of VIV depend on the time interval T. For the sake of clarity, this section focuses on data collected from hanger #1 in the 2021 period, during which it had no damper, resulting in measured responses that are considerably large. To ensure that mean wind direction does not affect the analysis, the data in this section has been limited to T-minute average wind direction from the sector 45° to 90°. This range is one of the most common directions at the site, which was shown in Fig. 9(a). This range was preferred over the symmetric 45° to 135° sector because, while being narrower, it only has 15% fewer data than the symmetric sector, and the discarded points had T-minute average wind magnitude mostly below 5 m/s.

As the time interval gets shorter, the number of calculated data points increases due to the finer time resolution. However, the number of compared points should be kept the same to ensure a fair comparison across different T values. The method utilized to achieve this result can be outlined using the 10-minute dataset, referred to as the $N_{10\text{min}}$ data set, and the 30-second dataset, the $N_{30\text{s}}$ data set, as an example. The objective is to create a new subset from the $N_{30\text{s}}$ set, having the same number of points as the $N_{10\text{min}}$ set. In this procedure, for each data point from $N_{10\text{min}}$ set, a corresponding point from $N_{30\text{s}}$ set, calculated from the same 10-minute period, is identified and selected. This results in a new data set, matching the number of points in the $N_{10\text{min}}$ set but consisting of data from the $N_{30\text{s}}$ set. In essence, this method allows for a direct and meaningful comparison between the two data sets, each originally averaged over different time scales. These points were then plotted against the average wind speed within the interval T, hereafter referred to as $\overline{U}$. This approach yielded an experimental representation of the velocity–amplitude (V–A) curve caused by VIV (Kim et al., 2022). Fig. 24 displays two examples of such scatter plots. A preliminary qualitative assessment suggests a noticeable similarity between the two data sets at the extremities of the T interval range. This implies that calculating the statistics over different T intervals seems to have a minor influence on the final shape of the V–A curve.

From an engineering perspective, the primary interest in analysing the VIV phenomenon lies in its impact on the service life of the hanger. A crucial metric in such analysis is the envelope of the standard deviation of displacement across varying wind velocities. A *backbone curve* is here defined as the 95th percentile of the standard deviation of the hanger displacement, conditioned upon the window averaged wind velocity $\overline{U}_j$. In practice, this conditional requirement is implemented by calculating the average wind velocity over bins called $\overline{U}_j$ of width $\Delta U = 0.8\,\text{m/s}$ with the index j indicating the left edge of the bin. Such a curve exists for each observation window T_i with $i \in \{30\text{ s}, 60\text{ s}, 5\text{ min}, 10\text{ min}\}$. It is therefore denoted as $X^i_{95}(\overline{U})$, or $X^i_{95,j} = X^i_{95}(\overline{U}_j)$.

Bootstrapping techniques are employed to analyse the two sets of points by generating 1000 random samples. Each sample consists of $N_{10}/2$ points, drawn from the respective sets. For every sample thus obtained, the 95th percentile backbone curve is computed. This process results in the creation of a statistical distribution for each wind speed bin $\overline{U}_j$, characterized by a mean $\overline{X}^i_{95,j}$ and a standard deviation $\sigma_{X^i_{95,j}}$. The aim of this methodological approach is to robustly estimate the statistical properties of the 95th percentile backbone curve for different wind speed conditions.

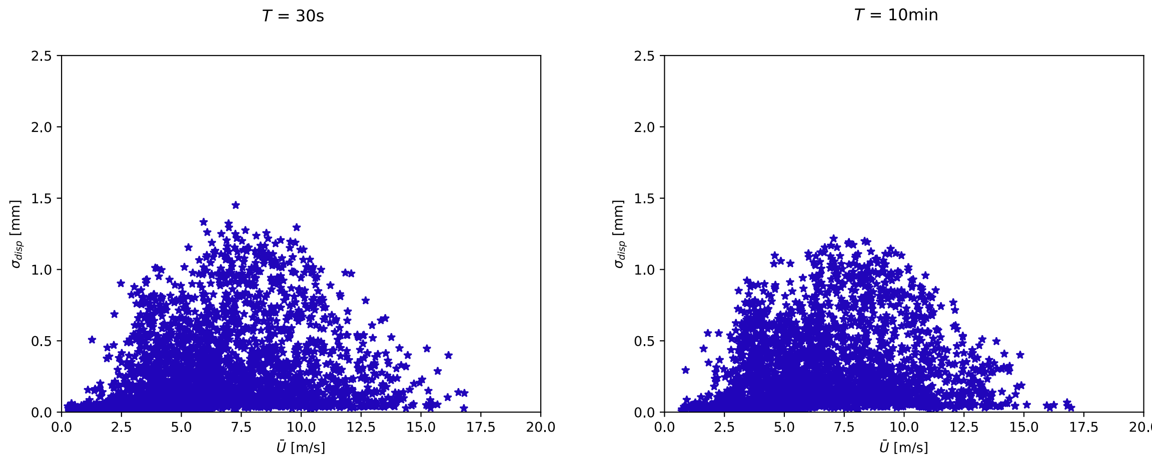
The dispersion of these 1000 envelope curves is illustrated in Fig. 25 for two observation windows. In both instances, the realization of the *backbone curve* provides a reliable estimate of the build-up to resonance and predicts the onset of full lock-in wind speed.

To quantitatively assess the differences between the *backbone curves*, the curves X^1_{95} and X^2_{95} in Fig. 25 are subjected to statistical hypothesis testing. The null hypothesis $\mathbb{H}_0$ posits that "*The means of* X^1_{95} *and* X^2_{95} *are equivalent.*" For a specific wind speed bin $\overline{U}_j$, the observed mean difference is $\hat{\Delta}_j = \left|\overline{X}^1_{95,j} - \overline{X}^2_{95,j}\right|$.

Under the assumption of $\mathbb{H}_0$, the probability of observing a difference as significant as $\hat{\Delta}_j$ is $P\left(\Delta_j > \left|\overline{X}^1_{95,j} - \overline{X}^2_{95,j}\right|\right)$. Assuming Δ_j follows a Gaussian distribution with zero mean and a standard deviation $\sigma_{\hat{\Delta}_j} = \left(\sigma^2_{X^1_{95,j}} + \sigma^2_{X^2_{95,j}}\right)^{0.5}$, the p-value for each wind bin $\overline{U}_j$ is computed as $p_j = 2\left[1 - \Phi\left(\hat{\Delta}_j/\sigma_{\hat{\Delta}_j}\right)\right]$, where Φ denotes the Gaussian cumulative distribution function. The significance level is set at $\alpha = 5\%$, and the p-value is calculated for each $\overline{U}_j$, comparing the smallest and largest T in the aforementioned range.

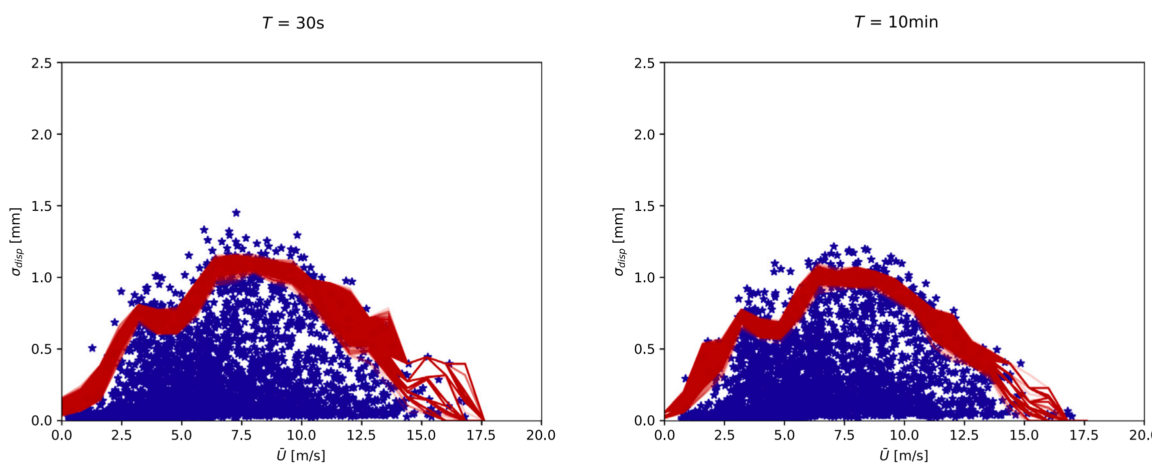
Should $p > \alpha$ for all $\overline{U}_j$, the two *backbone curves* are statistically indistinguishable, implying that the time interval T does not affect the determination of the response's backbone.

Fig. 26(a) presents three instances of the 95th percentile distribution for three distinct wind speed bins $\overline{U}_j \in \{3.2, 7.2, 12\}$ m/s. Notably,



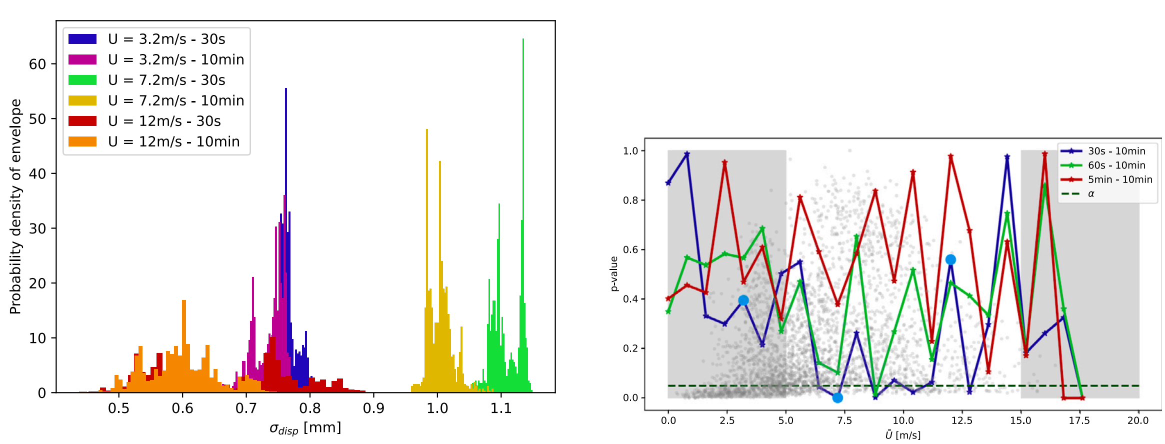
(a) (b)

Fig. 24. Standard deviation of the displacement on hanger #1 during 2021 (no damper) for two different time intervals T, filtered for wind direction $45° < \bar{\theta} < 90°$.



(a) (b)

Fig. 25. Standard deviation of the displacement on hanger #1 during 2021 (no damper) for two different time intervals T. In red the 1000 realizations of the backbone curve X^i_{95}, with (a) $i = 30$ s and (b) $i = 10$ min.



(a) (b)

Fig. 26. (a) Distribution of $X^i_{95_j}$ for $j = \{3.2$ m/s, 7.2 m/s, 12 m/s$\}$ and $i = \{30$ s, 10 min$\}$. (b) p-values for different wind speeds for three statistical hypothesis tests. Light blue dots indicate the configurations for which the distributions are detailed in Fig. 26(a). In the background, the point set is shown to guide the eye.

the standard deviation of the percentile gets larger with increasing wind speed, attributable to the scarcity of experimental data at higher velocities. The mean of the percentile distribution shows how the shorter time interval $T = 30$ s gives a higher displacement at 7.2 m/s.

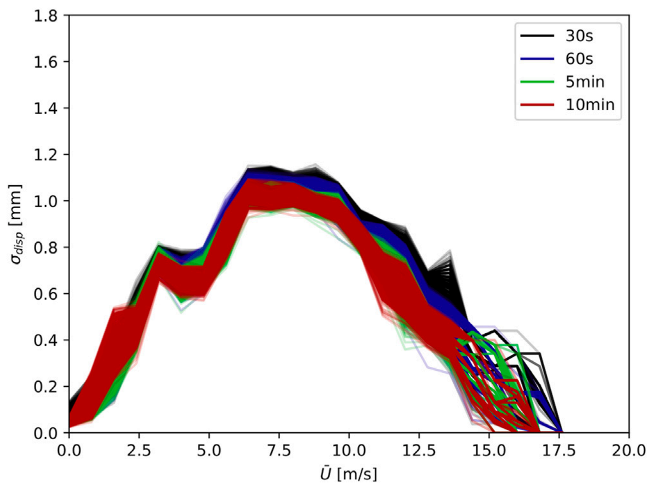


Fig. 27. Comparison of backbone realizations with different observation window length T_i.

This observation is substantiated in Fig. 26(b), which depicts the results of a statistical hypothesis test across three time interval combinations. In particular, the 10 min data set has been compared with the 30 s, 60 s and 5 min data sets. In the wind speed range where VIV is significant and sufficient experimental data exists (specifically between 5 m/s and 16 m/s), the *p*-value for each wind speed bin mostly exceeds the predefined significance level of the test when comparing 10 min with 60 s and 5 min. In particular, the closer the compared time interval lengths T are, the higher the *p*-value of the test, showing higher similarity between the two distributions. Consequently, the null hypothesis can be accepted for the red and the green comparison in Fig. 26(b), thereby confirming that in these cases the distributions X^1_{95} and X^2_{95} are statistically indistinguishable. However, it must be rejected for the comparison 30 s - 10 min, where the peak of the curve is reduced by about 9%. The same conclusion can be appreciated from Fig. 27 showing that, in general, the longer the window length T, the more the peak of the backbone is averaged down. This explains the failure of the test for the comparison between the shortest and the longest observation windows. From this analysis, it is inferred that within the wind speed range where VIV is evident for the hanger, the longer the length of the observation window T is chosen, the more the peak of the backbone curve is reduced. This suggests that the duration T should be chosen as short as possible to ensure a trustworthy estimation of the backbone. However, still considering the lower limits imposed by the timescales of the observed process.

4.3. Turbulence intensity effect on the response

In Fig. 20 it was shown the significant impact of turbulence intensity on the standard deviation statistics of the VIV response. Consequently, in conditions of high turbulence, the hanger does not reach the maximum displacement corresponding to the specific wind speed, primarily because high turbulence intensity tends to inhibit synchronization. To validate this hypothesis and further evaluate the method employed in the preceding section, a second statistical hypothesis test was conducted. This test compared VIV statistics derived from data sets with varying turbulence intensities. Fig. 10 illustrates the distribution of turbulence intensities within the collected data.

The data was divided into two groups based on crosswind turbulence intensity: $I_v < 0.15$ and $I_v > 0.15$. For each group, the response envelope was calculated using the 95th percentile method, as previously described, and is depicted in Fig. 28 for 1000 realizations.

Fig. 29(a) presents three instances of the 95th percentile distribution for different wind speeds $\overline{U}_j \in \{4, 7.2, 12\}$ m/s. Here, it is evident that the distributions look quite different for all three wind speeds. As the wind speed increases, the higher turbulence intensity group lacks sufficient data points to accurately estimate the backbone curve. Finally, Fig. 29(b) displays the results of the hypothesis test, indicating that for wind speeds between 5 m/s and 16 m/s, the null hypothesis must be rejected. This finding confirms that data sets with high turbulence intensity are characterized by a lower backbone curve compared to low turbulence data for the same wind speeds.

Combining the findings of sections Sections 4.2 and 4.3, it is found that the scatteredness in the V–A curves is better explained by wind turbulence than the possible difficulties related to the co-existence of several timescales in the statistical processing.

5. Conclusions

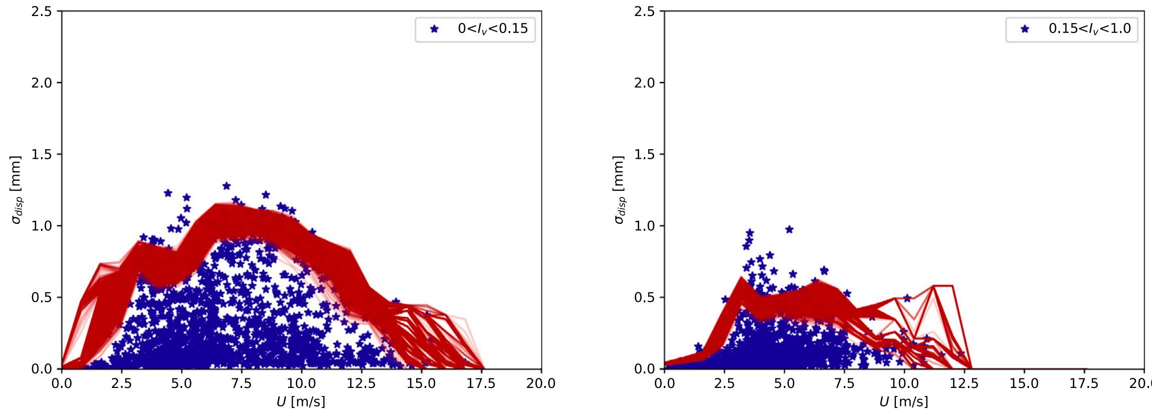
This study presents a statistical analysis of long-term hanger vibration data from the Hålogaland Bridge. The primary focus has been on evaluating the response of the bridge's three longest hangers, with findings indicating that their behaviour is predominantly caused by vortex-induced vibration. Key insights were gained by analysing time histories from a hanger without dampers, revealing that constant mean wind velocity and bounded wind direction are essential but not sufficient for a single-mode lock-in VIV. The process of entering and exiting lock-in is complex and appears to be influenced by factors beyond just wind speed and direction changes. The hangers' frequency spectrum is characterized by multi-frequency vibrations, frequently shifting around the Strouhal frequency, instead of a stable mono-harmonic state. Further, it is shown that installing one damper reduced VIV vibrations by 80%, but additional identical dampers, installed in the same spot, had a marginal effect and did not extend the range of frequencies damped by the devices. Nevertheless, the incorporation of extra dampers could potentially decrease vibrations of the dampers themselves, consequently extending their fatigue life. Turbulence intensity was found to significantly affect the maximum vibration amplitude during lock-in, with high turbulence disrupting vortex shedding and reducing maximum hanger displacement.

In the analysis of wind data, it was found that the slowly varying wind remains constant for less than three minutes in half of the observed instances. This is shorter than the typical time required for the complete development of lock-in vibrations. This characteristic is key to understanding why the distribution of the 10-minute averaged response vibration amplitudes across different wind speeds is highly skewed towards the smallest displacements, with occasionally larger peaks when the lock-in regime is reached.

A novel application of statistical hypothesis testing on experimental data has been used to investigate the influence of the duration of the time interval used to compute the statistical indicators. It has been shown that a longer time interval averages down the peak of the retrievable response envelope. However, the permissible length of the interval is also bounded below by the time scale of the observed phenomenon. Therefore, it is recommended to choose the shortest interval length that still guarantees a sufficient number of observed periods of the system under analysis.

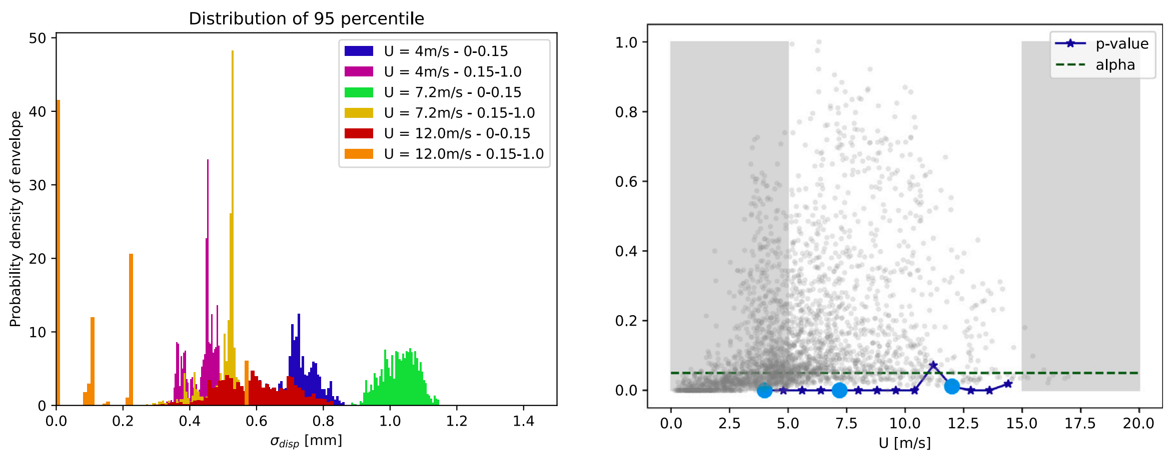
This technique has also confirmed turbulence's impact on the VIV response backbone curve. Higher turbulence intensities lead to a lower VIV amplitude. Therefore, the variability in the observed V–A curves is associated, mostly, with turbulence characteristics rather than potential challenges arising from the coexistence of multiple timescales in statistical processing. The presented methodology could be straightforwardly applied to the calculation of different backbones curves than the 95th one. It can be also used to test the influence of different external parameters on the capability to correctly reconstruct these curves from experimental data for different wind velocities.

This study highlights the discrepancies between the complex VIV behaviour observed in field data and the prevailing theoretical models of VIV. It emphasizes the need for broadening our knowledge about VIV phenomena, particularly in practical applications for long-span suspension bridges.



(a) (b)

Fig. 28. Standard deviation of the displacement on hanger #1 during 2021 (no damper) for two different crosswind turbulence intensity intervals. In red, the 1000 realizations of the backbone curve X^i_{95}.



(a) (b)

Fig. 29. (a) Distribution of $X^i_{95_j}$ for $j = \{4\ \text{m/s}, 7.2\ \text{m/s}, 12\ \text{m/s}\}$ and $i = \{0-0.15, 0.15-1\}$. (b) p-values for turbulence intensity comparison. In the background, the two sets of points are shown to guide the eye. Light blue dots indicate the configurations for which the distributions are detailed in Fig. 29(a).

CRediT authorship contribution statement

G. Bacci: Writing – original draft, Visualization, Software, Formal analysis, Data curation. **Ø.W. Petersen:** Writing – review & editing, Visualization, Supervision. **V. Denoël:** Supervision, Methodology, Conceptualization. **O. Øiseth:** Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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